

Incipient Fault Detection of Power Distribution System based on Statistical Characteristics and Transformer Network

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Abstract—Incipient fault detection has become increasingly important in power distribution systems since higher requirements on power supply reliability were proposed. With more sensors and developments of intelligent algorithms, data-driven methods are recognized as a promising direction. In this paper, two main parts of this time series classification task are discussed: feature extraction of events and fault detection classifiers. For feature extraction module, statistical characteristics are comprehensively analyzed and chosen with hypothesis test and FDR multiple test. For fault detection classifiers, one of the most recent and powerful models, Transformer Network, is introduced with corresponding adjustments for the fault detection application. The inside attention mechanism and ability to build long-distance data dependency are the reasons for the high-level performance of Transformer. Experiments prove that the proposed framework is reliable in different conditions and has much better performance than related works, which provides an excellent reference for time-series classification tasks in the field of online monitoring and fault diagnosis.

Keywords—Incipient fault detection; Power distribution systems; Statistical Characteristics; Transformer Network; Time-series classification

I. INTRODUCTION

With the rapid development of power systems, more requirements have been imposed on distribution network in terms of power supply reliability and economic benefits[1]. Distribution network has its own properties such as various types of equipment, complex network structure, wide distribution range and relatively low cost for equipment replacement. Therefore, post-fault repairing is widely adopted,

which brings the problem of low power supply reliability. With the gradually increased requirements from users, more and more studies are conducted on distribution network fault detection. Researches show that there exists a development process of equipment insulation deterioration where some specific kind of faults occur several times[2]. For this kind of faults, mechanisms are not the same in different types of power equipment in distribution system. However, they have common characteristics that are self-recovering and repeated occurrence. These properties are actually related to the state of equipment insulation. If the useful information can be effectively captured by engineers, it becomes possible to get early warnings for insulation failure, thus greatly reducing the error rate in distribution network.

Researchers in this area mainly focus on two kinds of frameworks: traditional mechanism-based method and data-driven method. Traditional models usually propose a simulation circuit to analyze arc-fault electric characteristics. Authors of Ref. [3] process fault waveform by modal analysis, and then built the transient circuit of the line by using the main mode. In Ref. [4], two different simulation systems are introduced for arc faults and non-arc faults respectively. Data-driven models mainly derive a discriminative function on fault categories from data. Ref. [5] proposed two incipient fault detection algorithms for the single-line-to-ground underground cables based on the methodology of wavelet analysis. Ref. [6] proposed an intelligent algorithm inspired by human perception and calculates fault probability with a generative framework. Ref.[7] builds a Siamese Temporal Graph to allow transfer learning between simulation data and field data. Ref.[8] designs an attention mechanism for waveform data processing. Although, there are some progresses made by previous

researchers. Both traditional models and data-drive models have their own limitations. Traditional models are not robust enough to adapt to various situations in distribution system, and data-drive models require a large number of training samples.

With the development of power distribution systems, it becomes more difficult to develop a physical circuit model for the complex system under various working conditions. On the other hand, increasing number of sensors have been installed in the system together with the large development of intelligent algorithms allow engineers to build powerful mathematical models to solve this fault detection problem. Although there have been a few works on mathematical model design, a systematical algorithm which explores both feature extraction and learning mechanism in details is lacked. In this paper, these two parts will be discussed with consideration of specific characteristics of incipient faults and property of different learning algorithms. The foundation of feature extraction is statistical characteristics which have been proven as effective strategy in pattern recognition[9], while in learning part the most recent and promising model, Transformer Network[10], is adopted. Experiments show that the proposed framework is reliable and has much better performance than related works.

II. FEATURE EXTRACTION OF EVENT

Appropriate and efficient feature extraction module is the key of incipient fault detection. For a dataset composed of N events, the j^{th} waveform from the i^{th} event is the a sequence $s_{i,j}(t_1), s_{i,j}(t_2), \dots, s_{i,j}(t_{n_i^{(j)}})$ where t is the sampling point and $n_i^{(j)}$ is the number of sampling points for j^{th} waveform. Then a matrix $S_{i,j} \in \mathbb{R}^{n_i^{(j)}}$ can be used to denote the sequence, that is

$$S_{i,j} = \begin{pmatrix} s_{i,j}(t_1), s_{i,j}(t_2), \dots, s_{i,j}(t_{n_i^{(j)}}) \end{pmatrix}^T \\ = \begin{pmatrix} s_{i,j,1}, s_{i,j,2}, \dots, s_{i,j,n_i^{(j)}} \end{pmatrix}^T \quad (1)$$

Where event is defined as a set of waveform records including three-phase voltage, three-phase current, neutral-point voltage and neutral-point current. The length of each type of waveform $n_i^{(j)}$ can be different.

To consider the periodic property of waveform, the feature extraction process is done cycle by cycle, shown in Fig.1. Since statistics are very useful information in incipient fault detection, our feature extraction is based on these statistical features. A general set of statistical features include:

- Summary statistics: maximum(S), minimum(S), mean(S), var(S), std(S), skewness(S), kurtosis(S), median(S), quantile of empiric distribution function (S|q).
- Additional characteristics: absolute energy(S), augmented dickey fuller test statistic(S), binned entropy(S|m), mass quantile(S|q).

- Dynamic features: Arima model coefficients(S|i,k), continuous wavelet transformation coefficients(S|a,b), fast fourier transformation coefficient(S|k).

Due to the large number of statistic indicators, feature engineering is conducive to removing interference and improving the data efficiency of model. For the above statistics, we calculate feature correlation, test importance and select them. For the classification problem, feature correlation can be expressed as whether there are significant differences in the distribution of the features from different categories, which can be judged by hypothesis test method. H_0 assumes that the feature is irrelevant to category prediction, and H_1 assumes the feature is related. According to the differences in feature types and sample categories, the p value can be obtained by using the accurate Fisher test, K-S test and Kendell rank test to determine the hypothesis. For the n_ϕ statistical indicators, the corresponding results are denoted by $(p_1, \dots, p_{n_\phi})^T$.

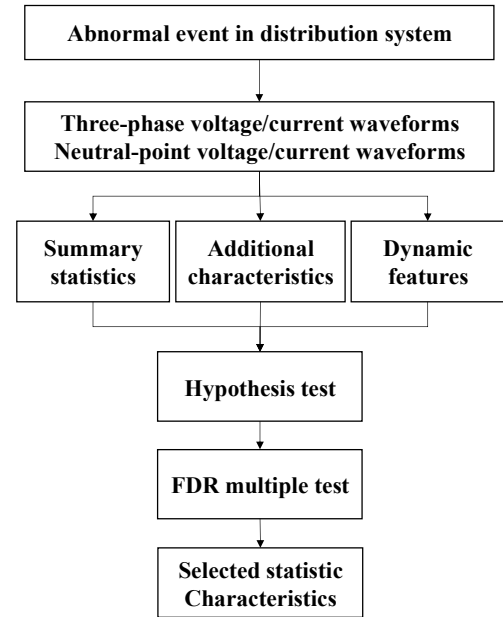


Fig. 1. The flowchart for feature extraction module

Due to the correlation between the statistics, FDR multiple tests are adopted: sorting the above p-value sequences, calculating the threshold r_ϕ according to the set value of false detection rate(FDR), and retaining the statistics whose p value is lower than the threshold.

$$r_\phi = \frac{\phi q}{n_\phi \sum_{\mu=1}^{\phi} \frac{1}{\mu}} \quad (2)$$

Where ϕ is the index of hypothesis sorted according to p value, q is the FDR rate, and n_ϕ is the total number of hypotheses.

III. DATA ANALYSIS

Since filed data are limited, simulation data are adopted to guarantee model performance comparison under various conditions. Besides, system design and parameter adjustment in simulation can help us familiarize incipient fault mechanism and therefore better analyze field data.

In PSCAD, we built a grounded 10 kV overhead line simulation system[11,12] which consists of six lines and is shown in Fig.2. Power frequency is set as 50 Hz with a 4k Hz sampling frequency. For each record, 16 cycles of three-phase voltage waveforms and current waveforms are captured and the trigger moment is fixed within the fourth cycle. Here f1,f2,f3 denote different fault locations, and d1,d2,d3 are corresponding distances. The fault simulation is based on Kizilcay arc model. L1,L2,L3 are resistance loads, and C1,C2,C3 are capacitive loads. These loads will randomly change as the simulation of transient disturbance(TD).

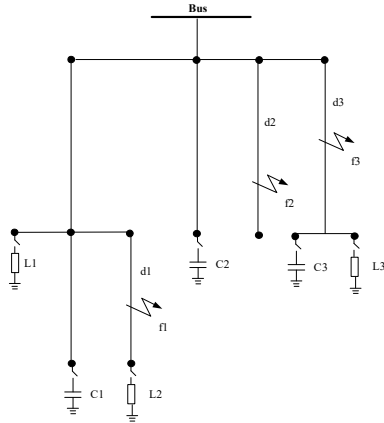


Fig. 2. The 10-kv-class power distribution simulation system

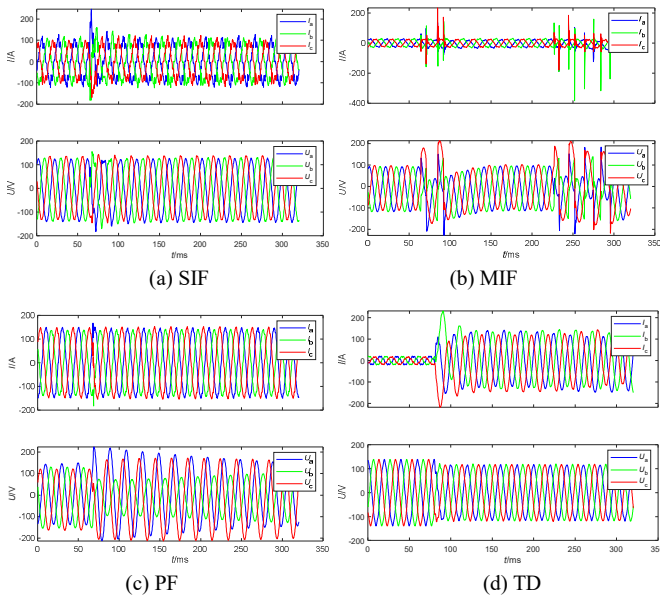


Fig. 3. Typical waveforms in field dataset

In addition to simulation data, field data from fault recording devices[13] are also used to provide an overall database. The systems are 10 kV-class overhead lines and all ungrounded. The form of field records is the same as simulation: 4 kHz sampling rate, 16 cycles of length and fourth cycle of trigger moments such as abnormal amplitude. Fig. 3 shows the typical waveform in field data. It can be seen that incipient fault is a self-recovering instantaneous fault. According to the fault duration, it can be divided into single-cycle incipient fault(SIF) and multi-cycle incipient fault(MIF), and the multi-cycle incipient faults indicate that the insulation state of the equipment is worse than that of single-cycle ones. The number of simulation data is 240, and that of field data is 198. The training-test split is 7:3. In simulation data, the distribution of four event types is 0.25:0.25:0.25:0.25. In field data, the distribution becomes 0.25:0.17:0.32:0.26. Based on field dataset, feature extraction module is applied. Some of the selected features (after normalization) for current waveform are shown in Fig.4. It can be concluded that these selected features have quite different distributions in the four event types. For example, maximum(S) of current waveform in SIF shows two clusters. One is around 1.8, and the other is focused on 2.2. This phenomenon indicates that the current waveform of SIF is always much higher than the normal value, and its exact value has two main tendencies. To contrast, the single cluster of maximum(S) in MIF is more distributed. In addition, from this index we can see that the maximum of current in TD is much lower than other three event types, which will be used in the following classifiers. Similar conclusions include: 1) The var(S) of current in MIF is higher than SIF; 2) The absolute energy(S) of current in TD is the lowest among all the four event types; 3) The 3rd harmonics attitude in FFT is always high in SIF, MIF, PF current waveform; 4) The detail attitude of current waveform in db4 wavelet transform has the following relative order: $SIF \approx MIF > PF \approx TD$.

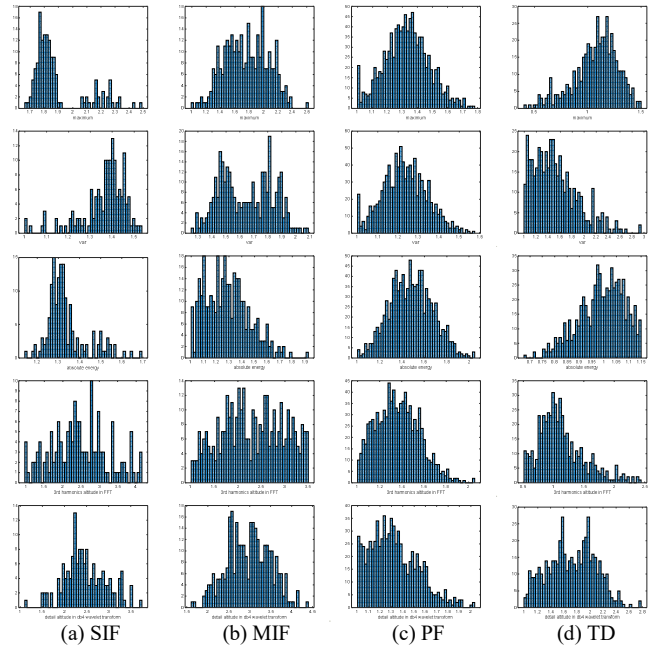


Fig. 4. Statistical characteristics distribution from different event types

IV. FAULT DETECTION BASED ON TRANSFORMER NETWORK

The statistical feature vectors for event have been extracted following the process illustrated in Section II. For the j^{th} waveform in i^{th} event, the l^{th} cycle is related to a set of filtered statistical indicators $s_{i,j,l}^{\phi}$. We combine them together into one feature vector as the representation for the l^{th} cycle of i^{th} event $u_l^i = [s_{i,1,l}^{\phi}, \dots, s_{i,j,l}^{\phi}, \dots, s_{i,8,l}^{\phi}]^T$. Now for i^{th} event, we have a set of feature vectors $\{u_1^i, \dots, u_l^i, \dots, u_L^i\}$ where L is equal to the number of cycles in a record.

The essence of the above problem is temporal correlation modeling, which is solved in this paper with Transformer Network. As one of most recent models, Transformer Network has been proved to be efficient and reliable. The central idea of this framework is attention mechanism, that is

$$f_{att}(Q, K, V) = f_{sfm} \left(\frac{QK^T}{\sqrt{d_k}} \right) V \quad (3)$$

Where Q, K, V are query vector, key vector and value vector. $\sqrt{d_k}$ is the scalar of normalization. f_{sfm} is the softmax function. f_{att} is the attention module.

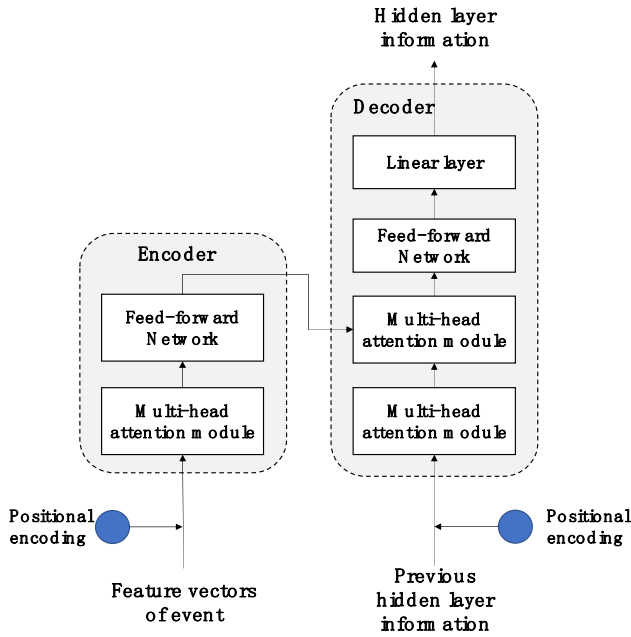


Fig. 5. The Framework of Transformer Network.

Based on this module, Transformer Network was proposed. First Q, K, V are calculated according to input vectors:

$$U^i = [(u_1^i)', \dots, (u_l^i)', \dots, (u_L^i)'] \quad (4)$$

$$(u_l^i)' = [u_l^i; p_l] \quad (5)$$

$$p_l^{(i)} := \begin{cases} \sin(\omega_k \cdot t), & \text{if } i = 2k \\ \cos(\omega_k \cdot t), & \text{if } i = 2k + 1 \end{cases} \quad (6)$$

$$\omega_k = \frac{1}{10000^{2k/d}} \quad (7)$$

$$\begin{bmatrix} Q^i \\ K^i \\ V^i \end{bmatrix} = \begin{bmatrix} W^Q \\ W^K \\ W^V \end{bmatrix} U^i$$

Where p_l is the positional encoding, ω_k is the frequency of sine functions, d is the encoding dimension, and W^Q, W^K, W^V are learnable parameter matrix. The attention mechanism can effectively mine the correlation between historical data and current data, so as to synthesize the two criteria and obtain the encoded features.

$$v^i = f_{m-h}(Q^i, K^i, V^i) \quad (8)$$

$$h^i = f_{FFN}(v^i) \quad (9)$$

$$= \max(0, v^i w_1 + b_1) w_2 + b_2$$

$$f_{masked-att} = f_{sfm} \left(W_{mask} \circ \frac{QK^T}{\sqrt{d_k}} \right) V \quad (10)$$

$$y^i = f_{sfm}(f_c(h_L^i)) \quad (11)$$

Where f_{FFN} is an element-wise non-linear mapping, w_1, w_2 are learnable weights, and b_1, b_2 are learnable offsets. $f_{masked-att}$ is the masked attention module used in decoder, and W_{mask} is the masked weighted matrix. h^i is the hidden layer information given by decoder, and h_L^i is the output of the last cycle L . f_c is the linear layer, and y^i is the predication of i^{th} event. The feature vectors processed by the above attention mechanism are further mapped by the fully connected feed-forward neural network, and the final coding results are obtained.

Decoding is the inverse process of encoding where the encoding results are processed by attention mechanism according to the hidden layer information of historical cycles. Different from the encoder, the decoder needs to consider two parts of data, which are reflected in the network structure, namely, the first layer input is the historical hidden layer information, the second layer input K, V vectors are from the encoding results, and the Q vector is from the first layer output. Finally, the prediction category is obtained by feed-forward network, linear layer and softmax function. The framework structure is shown in Fig. 5.

V. EXPERIMENTS

A. Results of the Proposed Algorithm

For the proposed method, the parameter values after adjustment are set as follows: FDR rate $q=0.1$, confidence $p=0.95$. The parameters of Transformer Network are set as: the number of layers in encoder and decoder are both 3, the number of heads of the multi-head attention mechanism module is 4, the model vector dimension is 128, the vector dimension of the full connected layer is 512, and the dropout rate[14] is 0.2. The model adopts Pytorch framework, the GPU type is GeForce RTX 2080 Ti, the optimizer is Adam, and the adaptive learning rate adopts pre-heating strategy, that is, the learning rate is low at the beginning and then increases gradually. After the pre-heating stage, the learning rate decreases exponentially, and the loss function adopts cross entropy.

The evaluation metric adopts F1 score[15]. Compared with precision rate, this index also considers recall rate. For conservative models, the precision rate can be improved by choosing simple samples and ignoring difficult samples. As the harmonic mean of precision and recall, F1 score will increase significantly only when both of them are at a high level. In addition, because F1 score is averaged among each category, the problem of data imbalance is alleviated.

TABLE 1 RESULTS OF THE PROPOSED METHOD

Type \ Model	Simulation		Field data	
	Tran	StatTran	Tran	StatTran
SIF	0.867	0.934	0.841	0.905
MIF	0.874	0.962	0.857	0.910
PF	0.840	0.951	0.834	0.924
TD	0.831	0.920	0.814	0.887
Ave.	0.853	0.942	0.837	0.907

To verify the performance improvement from statistical characteristics and Transformer Network, two models are considered: original waveform plus Transformer, and statistical characteristics plus Transformer. The results for the two models in both simulation data and field data are shown in Table 1, where Tran denotes the model of original waveform plus Transformer and StatTran denotes the model of statistical characteristics plus Transformer. It can be concluded that Transformer Network is suitable for all kinds of event types. This is because it can effectively model the dependency between different cycles, and therefore is good at time series data processing. On top of that, by using statistical characteristics, the model can better find the difference between different events and reach a higher accuracy. To be specific, the average accuracy of StatTran is 8.9% higher than that of Transformer network with simulation data. And for

field data, the average accuracy of StatTran is 7.0% higher than that of Transformer. These significant improvements show the advantages of using the proposed statistical characteristics. It can be concluded that the proposed statistical characteristics, compared with original waveform, are more powerful and data-efficient. This advantage is important for the incipient fault detection task of power distribution system. In addition, by analysing statistical characteristics, useful logical rules can be inferred with a combination of professional knowledges. Interpretability of AI models is a hot research topic for electric power industry where reliability is the first consideration.

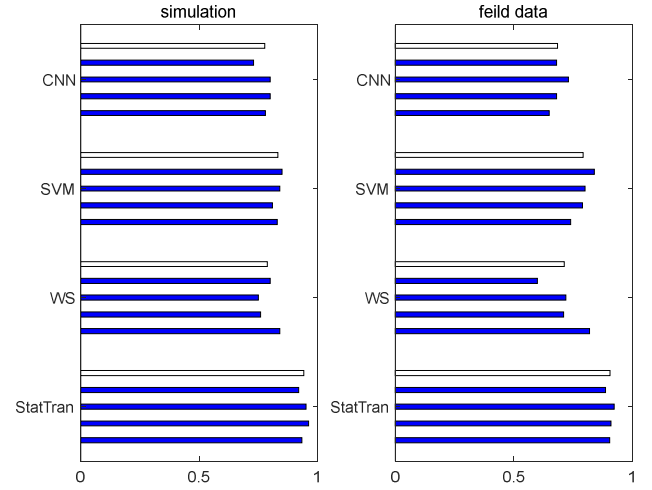


Fig. 6. Comparison with related works

B. Comparison with Related Works

To show the superiority of the proposed method, in this section it is compared with related works including Wavelet-based scheme(WS), support vector machine(SVM) and Convolutional Neural Network(CNN). The setting of WS is based on Ref. [7]. The input of SVM are statistical characteristics, and that of CNN are original waveforms. The results are shown in Fig. 6, where white bars denote average F1 score and blue bars denote F1 score of different event types. It can be concluded that the proposed method, StatTran, is better than other three methods both in simulation data and field data. This is because statistical characteristics can better describe the different event types than wavelet coefficients and original waveform. Besides, Transformer Network is better at processing time-series data than SVM and CNN. Therefore, the proposed method is proven as a good reference for time-series classification in power systems.

VI. CONCLUSION

In summary, the proposed method combines statistical characteristics with Transformer Network and is proven to be a promising framework in incipient fault detection task in distribution systems. On the other hand, interpretability and injection of prior knowledge are also very interesting topics, which will be discussed in future work.

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